Hao Li

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EDUCATION

The University of Manchester, UK 2025
Ph.D. in Computer Science

The University of Leeds, UK 2020
M.Sc. in Advanced Computer Science (AI)

Northeast Petroleum University, China 2017
B.Sc. in Information and Computing Science

RESEARCH DIRECTIONS

Foundation Model, Post-training and Alignment, Reinforcement Learning

WORK EXPERIENCE

ERNIE Team, Baidu Jan.2026 – Present
Researcher, China

- Technical Staff of the ERNIE Foundation Models Team, responsible for post-training and alignment of Baidu's in-house foundation model ERNIE, with a focus on improving and optimizing its general-purpose content generation capabilities and output quality.

Imperial College London Nov.2025 – Present
Postdoctoral Researcher, UK

- Led the research and prototyping of a differentially private synthetic data generation system for deployment within Trusted Research Environments (TREs).
- Collaborated with engineering teams to transform research code into containerised, production-ready software integrated into the TREvolution platform.

Microsoft Research Aug.2024 – May.2025
Research Internships, China

- Contributed to a **Foundation Model for Medical Time Series**, proposing Transformer improvements including **Continuous-Time RoPE** and **Frequency-based MoE**, and extending **Next-Token Prediction** to arbitrary time points via **Neural ODEs**, enhancing modeling of irregularly sampled clinical data.
- Core developer of the open-source project **TimeCraft** (**600+ GitHub stars** in three month), responsible for **Controllable Text-to-Time-Series Generation**:
  - Integrated **LLMs** with **Diffusion Models** to achieve cross-modal alignment and controllable generation;
  - Applied **Supervised Fine-Tuning (SFT)** across multi-domain datasets, significantly improving cross-domain and generalization performance.

NOTABLE PROJECT

TimeCraft – A Library for Controllable Time Series Synthesis Mar 2024 - Feb 2025

- Led the development of the **Text-to-Time-Series Generation Engine**, integrating **Diffusion Models** with **LLMs** to achieve cross-modal alignment and precise generation from natural language to complex time series;
- Proposed a **Synthetic Data Construction Framework**, leveraging **Multi-Agent** collaboration/adversarial mechanisms to generate high-quality training data, enabling large-scale controllable generation tasks;
- Promoted **Open-Source and Community Engagement**, with the project gaining **350+ GitHub stars** in its first month after release;

Awards & Funding

Research Grant (Medical Text Mining with Multi-task Learning) 2022
Leeds University Outstanding Student Scholarship 2019
First-class & Outstanding Graduate Scholarship, Northeast Petroleum University 2017
National First Prize, China College Big Data Competition 2016

PUBLICATIONS

Conference Proceedings

1. **Hao Li**, Yizheng Sun, Viktor Schlegel, Kailai Yang, Riza Batista-Navarro, and Goran Nenadic "Arg-LLaDA: Argument Summarization via Large Language Diffusion Models and Sufficiency-Aware Refinement", **Expected to ACL 2026** *(Meta Review:4, Review Score:3, 3.5, 4.5)*

2. **Hao Li**, Bowen Deng, Chang Xu, Zhiyuan Feng, Viktor Schlegel, Yu-Hao Huang, Yizheng Sun, Jingyuan Sun, Kailai Yang, Yiyao Yu, Jiang Bian. "MIRA: Medical Time Series Foundation Model for Real-World Health Data", **NeurIPS 2025** *(CCF-A, A*, 24.5% acceptance rate)*

3. **Hao Li**, Yuhao Huang, Chang Xu, Viktor Schlegel, Renhe Jiang, Riza Batista-Navarro, Goran Nenadic, Jiang Bian. "BRIDGE: Bootstrapping Text to Control Time-series Generation via Multi-agent Iterative Optimization and Diffusion Modelling", **ICML 2025** *(CCF-A, A*, 25.5% acceptance rate)*

4. **Hao Li**, Yuping Wu, Viktor Schlegel, Riza Batista-Navarro, Tharindu Madusanka, Iqra Zahid, Jiayan Zeng, Xiaochi Wang, Xinran He, Yizhi Li, Goran Nenadic. "Which Side Are You On? A Multi-task Dataset for End-to-End Argument Summarisation and Evaluation", **Findings of ACL 2024**

5. **Hao Li**, Viktor Schlegel, Riza Batista-Navarro, and Goran Nenadic. "Do You Hear The People Singing? Key Point Analysis via Iterative Clustering and Abstractive Summarisation", **ACL 2023** *(CCF-A, A*, 23.5% acceptance rate)*

6. Bowen Deng, Chang Xu, **Hao Li**, Yu-Hao Huang, Min Hou, Jiang Bian. "TarDiff: Target-Oriented Diffusion Guidance for Synthetic Electronic Health Record Time Series Generation", **KDD 2025** *(CCF-A, A*, 20.1% acceptance rate)*

7. Yizheng Sun, **Hao Li**, Chang Xu, Hongpeng Zhou, Chenghua Lin, Riza Batista-Navarro, Jingyuan Sun. "Does Acceleration Cause Hidden Instability in Vision Language Models? Uncovering Instance-Level Divergence Through a Large-Scale Empirical Study" , **EMNLP 2025***(CCF-B, A, 23.3% acceptance rate)*

8. Yizheng Sun, Qinshi Rui, **Hao Li**, Chenghua Lin, Riza Batista-Navarro. "LVPruning: An Effective yet Simple Language-Guided Vision Token Pruning Approach for Multi-modal Large Language Models", **Findings of NAACL 2025**

9. Yizhi Li, **Hao Li**, Chenghua Lin and 17 others. "CIF-Bench: The Challenging Chinese Instruction-Following Benchmark to Measure Generalizability of Large Language Models", **Findings of ACL 2024**

10. Tharindu Madusanka, Iqra Zahid, **Hao Li**, Ian Pratt-Hartmann and Riza Batista-Navarro. "Not All Quantifiers are Equal: Probing Transformer-based Language Models' Understanding of Generalised Quantifiers", **EMNLP 2023** *(CCF-B, A, 23.3% acceptance rate)*

Journal Articles

11. Yuanyuan Liang, Yanbing Ju, Xiao-Jun Zeng, **Hao Li**, Peiwu Dong, Tian Ju. "A user-generated content-based social network large-scale group decision-making approach in healthcare service: Case study of general practitioners selection in UK", **Expert Systems with Applications** *(Q1, SJR 1.875, IS 9.29)*

Extended Abstracts, Workshops, Posters

12. **Hao Li**, Yuping Wu, Viktor Schlegel, Riza Batista-Navarro, Thanh-Tung Nguyen, Abhinav Ramesh Kashyap, Xiaojun Zeng, Daniel Beck, Stefan Winkler, Goran Nenadic. "PULSAR: Pre-training with Extracted Healthcare Terms for Summarising Patients' Problems and Data Augmentation with Black-box Large Language Models", **BioNLP @ ACL 2023**

13. Poehl.Paul, **Hao Li**, Viktor Schlegel, Anil Bharath. "“Generating Realistic Multi-Beat ECG Signals", **DSP 2025**

14. Aishik Nagar, Viktor Schlegel, Thanh-Tung Nguyen, **Hao Li**, Yuping Wu, Kuluhan Binici, Stefan Winkler. "LLMs are not Zeroshot Reasoners for Biomedical Information Extraction", **Insights @ NAACL 2025**

15. Yizheng Sun, **Hao Li**, Riza Batista-Navarro. "LanViKD: Language-Vision Cross Modal Knowledge Distillation for Egocentric Action Recognition", **HAII @ ECAI 2024**

16. Viktor Schlegel, **Hao Li**, Yuping Wu, Anand Subramanian, Thanh-Tung Nguyen, Abhinav Ramesh Kashyap, Daniel Beck, Xiaojun Zeng, Riza Theresa Batista-Navarro, Stefan Winkler, Goran Nenadic. "Large Language Models Augmented by Synthetic Dialogue Convert Patient Dialogues to Medical Records", **Clef 2023 Working Note**

In Review & Preprint

17. **Hao Li**, Viktor Schlegel, Riza Batista-Navarro, and Goran Nenadic "Large Language Models in Argument Mining: A Survey", **Preprint**

18. Xu Zhang, Junwei Deng, Chang Xu, **Hao Li**, Jiang Bian "Continuous Time Series Generation with Irregular Observations", **Preprint**

19. Xu Zhang, **Hao Li**, Chang Xu, Yuhao Huang, Jiang Bian "A Survey on Time-Series Generation: Taxonomy, Review and Prospects", **Preprint**

20. Kailai Yang, Xiao Liu, Lei Ji, **Hao Li**, Yeyun Gong, Peng Cheng, Mao Yang. "Data Mixing Agent: Learning to Re-weight Domains for Continual Pre-training", **Preprint**

21. Yidan Sun, Viktor Schlegel, Srinivasan Nandakumar, Iqra Zahid, Yuping Wu, Yulong Wu, **Hao Li**, Jie Zhang, Warren Del-Pinto, Goran Nenadic, Siew Kei Lam, Anil Anthony Bharath. "SYNBENCH: A Benchmark for Differentially Private Text Generation", **Preprint**

22. Yuping Wu, Viktor Schlegel, Warren Del-Pinto, Srinivasan Nandakumar, Iqra Zahid, Yidan Sun, **Hao Li**, Usama Farghaly Omar, Amirah Jasmine, Arun-Kumar Kaliya-Perumal, Chun Shen Tham, Gabriel Connors, Anil Anthony Bharath, Goran Nenadic. "Term2Note: Synthesising Differentially Private Clinical Notes from Medical Terms", **Preprint**

23. Yuping Wu, **Hao Li**, Hongbo Zhu, Goran Nenadic, Xiao-Jun Zeng. "Unifying Extractive and Abstractive Summarization within the Single Encoder-Decoder Framework", **Preprint**